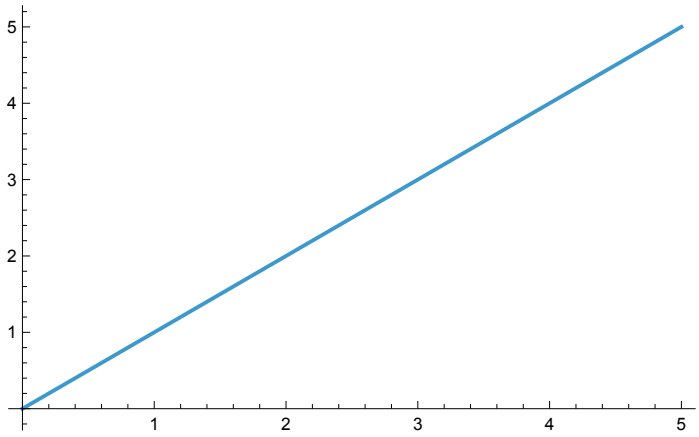
Problem 1

In[ ]:=
```
Plot[x, {x, 0, 5}]
```

Out[ ]=




Problem 2

In[ ]:=
```
D[f[t], t]
D[f[t], {t, 2}]
f[t_] := α Log[a t] Cos[ω t] Sin[ω t]
```

Out[ ]=
```
f'[t]
```

Out[ ]=
```
f''[t]
```

Problem 3

In[ ]:=
```
Clear[f]

w[x_] := f[x] × g[x] × h[x]
D[w[x], x]
```

Out[ ]=
```
g[x] × h[x] f'[x] + f[x] × h[x] g'[x] + f[x] × g[x] h'[x]
```

Problem 4

In[ ]:=

```
Clear[t, ϕ]
g = 9.81;
l = 10;
ω = Sqrt[g / l];

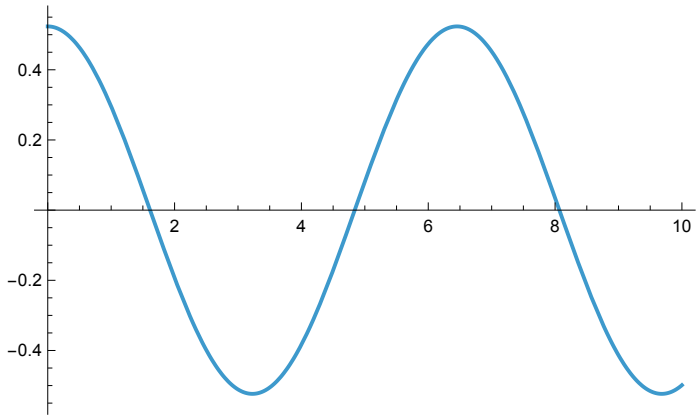
sol = NDSolve[{ϕ''[t] == -ω^2 Sin[ϕ[t]], ϕ'[0] == 0, ϕ[0] == 30 Degree}, ϕ, {t, 0, 10}]
Plot[ϕ[x] /. sol, {x, 0, 10}]
```

Out[ ]=

$$\left\{\left\{\phi \to \text{InterpolatingFunction}\left[\right]\right\}\right\}$$

Domain: {{0., 10. }}
Output: scalar

Out[ ]=




Problem 5

In[ ]:=

```
Graphics3D[
 {Cylinder[{{0, 0, 0.25}, {0, 0, 6}}, 0.3], Cylinder[{{0, 0, 0}, {0, 0, 0.25}}]}]
```

Out[ ]=

